Well Ordering and the Induction Principle

- **Exercise 1**
  Consider $A = \{n \in \mathbb{Z} : n > 10\}$.
    1. Give 5 different lower bounds of A.
    2. Which of those lower bounds belong to A?
    3. Find $\min A$.
    4. Explain why 10 is not the minimum.
    - **Solution 1.1**
        1. $10, 9, 8, 7, 6$.
        2. None.
        3. As A is bounded below, $\min A$ exists. Since $n \geqslant 10 + 1 = 11$, 11 is a lower bound of A. Since $11 \in A$, we have $11 = \min A$.
        4. $10 \notin A$.

- **Exercise 2**
  Prove that $\forall n \in \mathbb{N}, 1 + 3 + \cdots + (2n - 1) = n^2$.
    - **Proof 2.1**
      By induction.
        1. Base step
           If $n = 1$ the result is true because $1 = 1^2$.
        2. Inductive step
           Suppose that the result is true for n, that is
           $$1 + 3 + \cdots + (2n - 1) = n^2.$$
           Hence
           $$\begin{aligned} 1 + 3 + \cdots + (2n - 1) + (2(n + 1) - 1) &= (1 + 3 + \cdots + (2n - 1)) + (2n + 1) \\ &= n^2 + 2n + 1 \\ &= (n + 1)^2. \end{aligned}$$
           Then the result is true for $n + 1$ and therefore
           $$1 + 3 + \cdots + (2n - 1) = n^2,$$
           is true for every $n \in \mathbb{N}$.

- **Exercise 3**
  Prove that $\forall n \in \mathbb{N}, 2^n \geqslant n + 1$.
    - **Proof 3.1**
      By induction.
        1. Base step
           If $n = 1$ the result is true because $2^1 \geqslant 1 + 1$.
        2. Inductive step
           Suppose that the result is true for n, that is

$$2^n \geqslant n+1.$$

Because $n \geqslant 1$, $2n \geqslant n$, hence

$$2^{n+1} = 2 \cdot 2^n \geqslant 2 \cdot (n+1) = 2n + 2 \geqslant (n+1) + 1.$$

Then the result is true for $n+1$ and therefore

$$2^n \geqslant n+1,$$

is true for every $n \in \mathbb{N}$.

- **Exercise 4**
  Prove that $\forall n \in \mathbb{N}, n \geqslant 4, 2^n \geqslant n^2$.
  - **Proof 4.1**
    By induction.
    1. Base step
       If $n = 4$ the result is true since $2^4 = 16 \geqslant 16 = 4^2$.
    2. Inductive step
       Suppose that the result is true for $n \geqslant 4$, that is

       $$2^n \geqslant n^2.$$

       Hence

       $$\begin{aligned} 2^{n+1} &= 2 \cdot 2^n \\ &\geqslant 2n^2 \\ &= n^2 + n^2. \end{aligned}$$

       Since $n \geqslant 4$, we have that $n^2 \geqslant 4n = 2n + 2n > 2n + 1$. Therefore

       $$2^{n+1} \geqslant n^2 + 2n + 1 = (n+1)^2.$$

       Then the result is true for $n+1$ and therefore

       $$2^n \geqslant n^2,$$

       is true for every $n \in \mathbb{N}$ with $n \geqslant 4$.

- **Exercise 5**
  Prove that $\forall n \in \mathbb{N}, 3 \mid (4^n - 1)$.
  - **Proof 5.1**
    By induction.
    1. Base step
       If $n = 1$ the result is true since $3 \mid (4^1 - 1)$.
    2. Inductive step
       Suppose that the result is true for n, that is

       $$3 \mid (4^n - 1)\,.$$

       By definition, $\exists k \in \mathbb{Z}, 4^n - 1 = 3k \implies 4^n = 3k + 1$. Then

$$\begin{aligned} 4^{n+1} - 1 &= 4 \cdot 4^n - 1 \\ &= 4 \cdot (3k+1) - 1 \\ &= 12k + 3 \\ &= 3 \cdot (4k+1). \end{aligned}$$

Since $4k+1 \in \mathbb{Z}$, $3 \mid (4^{n+1} - 1)$. Then the result is true for $n+1$ and therefore

$$3 \mid (4^n - 1)$$

is true for every $n \in \mathbb{N}$.

- **Exercise 6 (Fibonacci Inequality)**
  Prove that $\forall n \in \mathbb{N}, F_n < 2^n$ where F_n is defined by $F_1 = 1, F_2 = 1, F_{n+2} = F_{n+1} + F_n$.
    - **Proof 6.1**
      By induction.
        1. Base step
           If $n = 1, 2$ the result is true since $F_1 = 1 < 2 = 2^1, F_2 = 1 < 4 = 2^2$.
        2. Inductive step
           Suppose that the result is true for both n and $n+1$, that is

           $$F_n < 2^n \wedge F_{n+1} < 2^{n+1}.$$

           By definition

           $$\begin{aligned} F_{n+2} &= F_{n+1} + F_n \\ &< 2^{n+1} + 2^n \\ &< 2^{n+2}. \end{aligned}$$

           Then the result is true for $n+2$ and therefore

           $$F_n < 2^n,$$

           is true for every $n \in \mathbb{N}$.

- **Exercise 7 (Intersection of lines)**
  Suppose that n lines are drawn in the plane so that
    1. no two lines are parallel;
    2. no three lines pass through the same point.

  Prove that the number of intersection points is $\dfrac{n(n-1)}{2}$.
    - **Proof 7.1**
      By induction.
        1. Base step
           If $n = 1$ the result is true since the number of intersection points is $0 = \dfrac{1(1-1)}{2}$.
        2. Inductive step
           Suppose that the result is true for n, that is, any n lines satisfying the conditions have

           $$\frac{n(n-1)}{2}$$

           intersection points.

Now consider the case with $n+1$ lines. When we add the $(n+1)$-th line to the existing n lines:

- Since no two lines are parallel, the new line must intersect with all of the existing n lines, creating n new intersection points.
- Since no three lines pass through the same point, all of these n new intersection points are distinct from the previous ones.

Hence, the total number of intersection points for $n+1$ lines is

$$\begin{aligned}\frac{n(n-1)}{2}+n &= \frac{n(n-1)+2n}{2}\\ &= \frac{n^2-n+2n}{2}\\ &= \frac{n^2+n}{2}\\ &= \frac{(n+1)n}{2}.\end{aligned}$$

Then the result is true for $n+1$ and therefore the number of intersection points is $\dfrac{n(n-1)}{2}$ for every $n \in \mathbb{N}$.